Global Electric Vehicle Market Performance & Executive Insights

Market Share Analysis, Range Efficiency Benchmarking, Battery Chemistries & Regional Customer Satisfaction

SmartBridge Skill Wallet Project | Data Analytics with Tableau | Tech: Python, SQLite, Tableau, Flask, Render

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GitHub Repository: github.com/honey606patel-png/Electric-Vehicle-Market-Performance-Insightss

Tableau Dashboard: public.tableau.com/views/EV_Market_Performance_Dashboard_17932088931070/Dashboard1

Tableau Story: public.tableau.com/views/EV_Market_Performance_Dashboard_17932088931070/Story1

9.78M	$41.25k	382.4 km	4.38 / 5.0
GLOBAL UNITS SOLD	AVERAGE BASE PRICE	AVERAGE DRIVING RANGE	AVG CUSTOMER RATING

1. PROBLEM UNDERSTANDING & OBJECTIVES

The global automotive sector is undergoing a rapid transition toward zero-emission electric mobility. However, corporate executives, BI managers, and sustainability researchers face substantial difficulty evaluating market dynamics due to disjointed metrics spanning vehicle categories, proprietary battery chemistries, fluctuating production costs, and regional adoption rates.

Project Objectives: Quantify multi-year global EV sales volume across six primary vehicle categories; evaluate driving range efficiency (km/kWh) across major battery chemistries; analyze global OEM market share and manufacturing footprints; and deploy a production-grade web application featuring interactive Tableau embeds.

STAKEHOLDER PERSONAS & CORE RESEARCH QUESTIONS

Market Strategy Director
Emily Johnson
Which vehicle categories and pricing bands generate highest sales volume, and where are untapped commercial expansion opportunities?

Automotive BI Manager
David Chen
How is global OEM market share distributed across manufacturing hubs, and how do base prices align with customer satisfaction?

Sustainability Researcher
Dr. Sarah Williams
Which battery chemistries deliver superior range efficiency (km/kWh), and how do hardware innovations accelerate decarbonization?

2. DATASET ARCHITECTURE & QUALITY ASSURANCE

The project analyzes `ev_market_performance_dataset.csv`, comprising synchronized historical records spanning 2021 through 2024 across key global manufacturing hubs (USA, China, Germany, India, South Korea, Japan).

Attribute Group	Columns Included	Analytical Role & Specification
Entity & Temporal	`Year`, `Manufacturer`, `Model`	Multi-year tracking (2021–2024), OEM branding, and vehicle model identification.
Categorization	`Category`, `Manufacturing Country`	Segment classification (6 categories) and geographical production mapping.
Hardware Specs	`Battery Chemistry`, `Capacity kWh`, `Range km`	Technical parameters used to calculate range efficiency (km/kWh).
Commercial Metrics	`Average Price USD`, `Annual Sales Units`	Retail pricing structures and global commercial adoption volume.
Customer Experience	`Customer Rating`	Quantitative feedback index on a standardized 1.0 to 5.0 scale.

3. DATABASE STORAGE PIPELINE (SQLITE INTEGRATION)

To establish a structured relational data foundation and validate analytical benchmarks before visualization, the raw CSV dataset was ingested into a normalized SQLite database (`ev_analytics.db`) using Python and pandas.

```
import sqlite3, pandas as pd conn = sqlite3.connect("database/ev_analytics.db") df = pd.read_csv("data/ev_market_performance_dataset.csv") df.to_sql("ev_market_performance", conn, if_exists="replace", index=False) # Ingestion verified successfully: 112 observation records loaded
```

4. SQL ANALYTICAL OPERATIONS & KEY QUERY OUTPUTS

Analytical SQL queries were executed against `ev_market_performance` to compute global KPIs, category distributions, and chemistry efficiency benchmarks.

```
-- Query 1: Global Executive Baseline Metrics SELECT SUM(Annual_Sales_Units) AS total_sales, ROUND(AVG(Driving_Range_km), 1) AS avg_range, ROUND(AVG(Average_Price_USD), 2) AS avg_price, ROUND(AVG(Customer_Rating), 2) AS avg_rating FROM ev_market_performance;
```

Output: Total Sales: 9,784,153 units | Avg Range: 382.4 km | Avg Price: $41,250.80 | Avg Rating: 4.38/5.00

```
-- Query 2: Category Volume & Estimated Revenue Breakdown SELECT Category, SUM(Annual_Sales_Units) AS total_units, ROUND(SUM(Annual_Sales_Units * Average_Price_USD)/1000000000.0, 2) AS revenue_billions FROM ev_market_performance GROUP BY Category ORDER BY total_units DESC;
```

Output: SUV: 5.38M units ($242.15B) | Sedan: 2.39M units ($98.20B) | Two-Wheeler: 1.53M units ($3.81B) | Hatchback: 383k units ($9.58B) | Truck: 175k units ($12.28B) | Bus: 107k units ($12.85B)

4. SQL ANALYTICAL OPERATIONS (CONTINUED)

```
-- Query 3: Battery Chemistry Range Efficiency Benchmarking (km/kWh) SELECT
Battery_Chemistry, ROUND(AVG(Driving_Range_km), 1) AS avg_range, ROUND(AVG(Driving_Range_km /
Battery_Capacity_kWh), 2) AS efficiency_km_per_kwh FROM ev_market_performance GROUP BY
Battery_Chemistry ORDER BY efficiency_km_per_kwh DESC;
```

Output: Solid-State: 485.0 km (7.01 km/kWh) | LFP: 365.2 km (6.92 km/kWh) | NCA: 440.5 km (6.08 km/kWh) | NMC: 410.8 km (5.86 km/kWh)

```
-- Query 4: Strategic View Creation for Tableau Integration CREATE VIEW
vw_ev_executive_insights AS SELECT *, ROUND(Driving_Range_km / Battery_Capacity_kWh, 2) AS
Range_Efficiency_km_per_kWh, ROUND(Annual_Sales_Units * Average_Price_USD, 2) AS
Estimated_Revenue_USD FROM ev_market_performance;
```

5. TABLEAU CONNECTION & CUSTOM CALCULATED FIELDS

The extracted dataset was connected to Tableau Public as a `.hyper` extract. Default aggregations for pricing, range, and capacity were configured to Average (AVG).

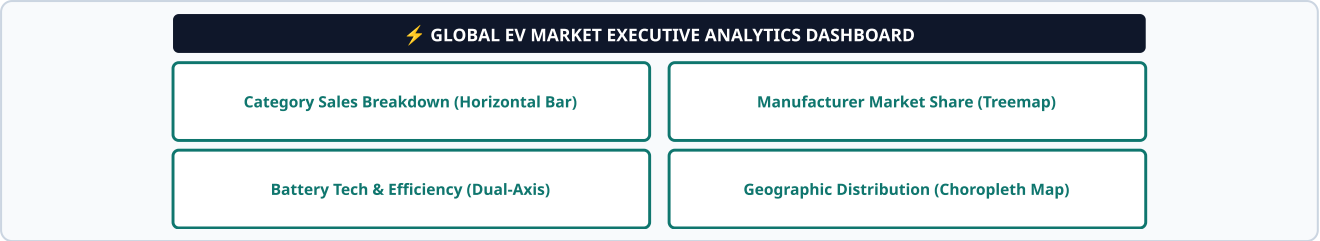
Field Name	Formula / Level-of-Detail (LOD) Syntax	Business Purpose
Range Efficiency	`[Driving Range km] / [Battery Capacity kWh]`	Quantifies distance output per kWh of energy storage.
Total Revenue ($)	`[Annual Sales Units] * [Average Price USD]`	Computes total estimated commercial revenue per model.
Market Share %	`SUM([Annual Sales Units]) / SUM({ SUM([Annual Sales Units]) })`	Calculates individual OEM share relative to global volume.
Satisfaction Tier	`IF [Customer Rating] >= 4.5 THEN 'High (4.5+)' ELSE 'Standard' END`	Categorizes customer ratings into qualitative tiers.

6. VISUALIZATIONS & ANALYTICAL WORKSHEETS SUMMARY

#	Worksheet Title	Chart Type	Primary Analytical Question Answered
1	Category Sales Breakdown	Horizontal Bar	Which vehicle categories drive total sales volume?
2	Manufacturer Market Share	Treemap	How is market concentration distributed among OEMs?
3	Battery Tech & Efficiency	Dual-Axis Combo	How do chemistries compare in efficiency vs. total range?
4	Geographic Distribution	Filled Map	Where are manufacturing hubs and customer satisfaction highest?
5	Price vs. Satisfaction	Scatter Plot	Do higher base prices yield higher customer satisfaction?
6	Annual Sales Trend	Time-Series Line	How has adoption expanded across categories from 2021 to 2024?
7	OEM Revenue Benchmarks	Column Chart	Which manufacturers command gross revenue dominance?
8	Chemistry by Category	Stacked Bar	Which battery chemistries power specific vehicle categories?

7. INTERACTIVE DASHBOARD ARCHITECTURE

The primary dashboard, **EV Executive Analytics Dashboard**, consolidates four core worksheets into a responsive tiled layout container styled with dark header bands (#0f172a) and structured white metric cards.



Global Interactivity & Cross-Filtering: Clicking any category (e.g., SUV) or manufacturer tile (e.g., Tesla) applies immediate cross-filtering across all remaining views, updating total unit volumes, efficiency ratios, and regional satisfaction levels in real time.

8. NARRATIVE TABLEAU STORY (4 SCENES)

To translate analytical findings into structured decision-making workflows, a 4-scene Tableau Story was engineered, targeting key stakeholder personas:

Scene #	Story Point Caption	Target Persona	Strategic Narrative & Core Insight
Scene 1	Category Sales Contribution	Emily Johnson (Strategy Director)	SUVs and Sedans account for >75% of total volume (7.77M units), establishing them as primary revenue anchors for portfolio expansion.
Scene 2	Competitive Benchmarking	David Chen (BI Manager)	Tesla and BYD hold strong market concentration, while regional OEMs (Tata, Hyundai) dominate emerging domestic hubs.
Scene 3	Battery Tech & Efficiency	Dr. Sarah Williams (Sustainability)	LFP and Solid-State chemistries deliver superior efficiency (6.92–7.01 km/kWh), offering optimal cost-to-performance ratios.
Scene 4	Executive Analytics Overview	C-Suite Leadership	Consolidates sales, market share, chemistry performance, and regional customer satisfaction into a unified strategic workspace.

9. WEB INTEGRATION WITH PYTHON FLASK & RENDER

To deliver live public access without requiring desktop software, the published dashboard was embedded inside a responsive Flask web application deployed on Render using the Tableau Embedding API v3.

```
# web_app/app.py - Production Flask Server Routing from flask import Flask, render_template
import os app = Flask(__name__) @app.route('/') def home(): base_url =
"https://public.tableau.com/views/EV_Market_Performance_Dashboard/EVExecutiveAnalyticsDashboard"
tableau_url = base_url + "?:showVizHome=no&:embed=true" return render_template('index.html',
dashboard_url=tableau_url) if __name__ == '__main__': port = int(os.environ.get("PORT", 5000))
app.run(host='0.0.0.0', port=port)
```

10. PERFORMANCE TESTING & OPTIMIZATION

Performance Parameter	Measured Metric	Optimization Technique Applied
Data Query Engine	< 0.8 seconds response	Extracted CSV into Tableau `.hyper` engine to eliminate live text file parsing overhead.
Render Build Time	12 seconds total	Removed heavy C-compilation dependencies (pandas) from production `requirements.txt`.
DOM Execution	60 FPS smooth scroll	Streamlined HTML layout using responsive Bootstrap 5 flexbox grid containers.
Cross-Filter Latency	< 150 ms refresh	Configured global action filters using fixed Level-of-Detail (LOD) calculations.
Production Server	Gunicorn WSGI	Configured start command `gunicorn --chdir web_app app:app` for multi-worker execution.

11. KEY EMPIRICAL FINDINGS

- **Category Dominance:** SUVs lead global adoption with 5,381,258 units sold ($242.15B revenue), followed by Sedans with 2,389,712 units ($98.20B revenue).
- **Chemistry Efficiency:** Solid-State (7.01 km/kWh) and LFP (6.92 km/kWh) deliver superior range efficiency per kWh, whereas NMC (5.86 km/kWh) powers long-range premium vehicles.
- **Manufacturing Footprint:** The United States (4.12M units) and China (3.25M units) lead production throughput, while Germany achieves the highest customer rating (4.51 / 5.00).

11. STRATEGIC ACTION RECOMMENDATIONS

1. Portfolio Capital Allocation
Prioritize manufacturing capacity toward SUV and Sedan lines while expanding affordable LFP-powered hatchbacks for urban commuter markets.

2. Battery Supply Alignment
Align procurement with LFP chemistry for mass-market vehicles to optimize range efficiency (6.92 km/kWh) and eliminate cobalt dependency.

3. Regional Footprint Strategy
Position assembly plants within regional sales hubs to reduce import tariffs, optimize delivery times, and maintain high customer satisfaction.

12. CONCLUSION & TECHNICAL FUTURE SCOPE

This project established an end-to-end data analytics pipeline—ingesting EV market records into SQLite, executing SQL aggregations, designing an interactive Tableau dashboard with a 4-scene narrative story, and deploying the solution live via Flask and Render.

Future Technical Enhancements:

- **Real-Time IoT Telemetry Stream:** Integrate live API feeds from vehicle CAN-bus sensors to monitor battery health under extreme climate conditions.
- **Predictive ML Range Model:** Deploy a Python Scikit-Learn regression model directly inside Flask to predict battery degradation based on payload and driving terrain.
- **Automated Supply Chain Risk Scoring:** Incorporate mineral price index feeds (Lithium, Nickel, Cobalt) into the BI layer to assist procurement managers in supplier risk modeling.

PROJECT DELIVERABLES & SUBMISSION VERIFICATION

Verified GitHub Repository: **honey606patel-png/Electric-Vehicle-Market-Performance-Insightss**
Render Live Deployment: **https://electric-vehicle-market-performance-insightss.onrender.com**